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RBE211TC DYNAMIC SYSTEMS

RESPONSES OF LINEAR DYNAMIC SYSTEMS

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Lecture Outline

1. Brief recap of first-order system response
2. Response of second-order systems
3. Interpretation of natural frequency, damping ratio, oscillation and decay
4. Link to damping-ratio estimation

First-Order Systems: A Brief Revisit

You have already seen first-order systems in:

- **RL circuits**
- **RC circuits**

Key features:

- one energy-storage element
- one time constant
- exponential transient response
- no oscillation

First-order systems provide a useful contrast to **second-order systems**.

Natural Response of First-Order Systems

For a first-order system, the natural response typically takes the form:

$$x(t) = x(0)e^{-t/\tau}$$

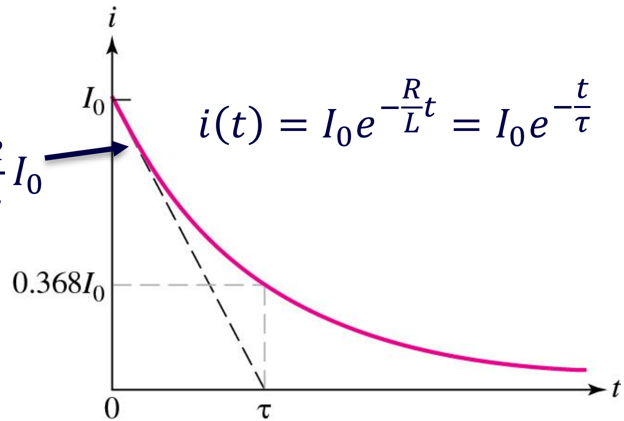
where

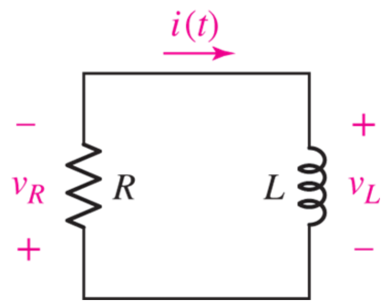
$x(0)$: initial value

τ: time constant

Slope: $-\frac{R}{L}I_0$

$$i(t) = I_0 e^{-\frac{R}{L}t} = I_0 e^{-\frac{t}{\tau}}$$





Example: Natural response of an RL circuit.

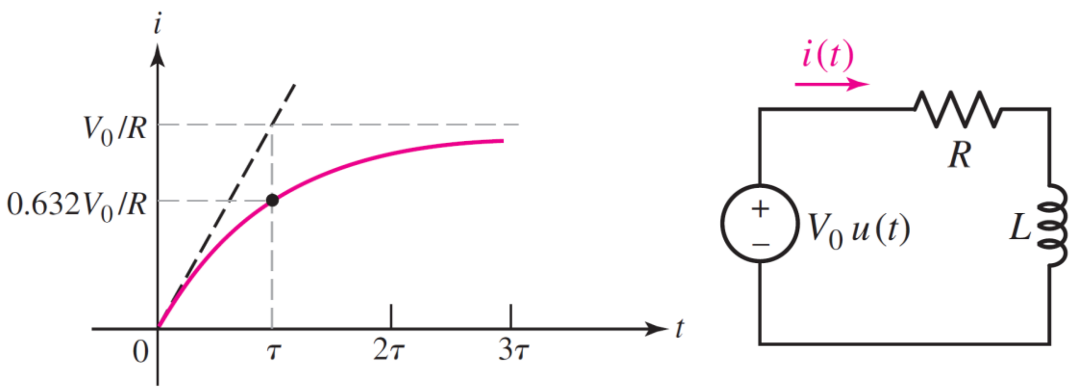
Step Response of First-Order Systems

The step response of a first-order system typically takes the form:

$$x(t) = x(\infty) + [x(0) - x(\infty)]e^{-t/\tau}$$

where

$x(\infty)$: steady-state value



Example: Step response of an RL circuit.

Transition to Second-Order Systems

First-order systems:

- one time constant
- exponential response
- no oscillation

Second-order systems introduce:

- **two characteristic roots**
- **natural frequency** ω_n
- **damping ratio** ζ
- possible **oscillation**, **overshoot**, and non-monotonic **decay**

Standard Form of a Second-Order System

A general second-order linear system may be written as

$$a_2\ddot{x} + a_1\dot{x} + a_0x = f(t)$$

A common normalized form is

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = b\,u(t)$$

where:

ω_n: natural frequency

ζ: damping ratio

Free Response of a Second-Order System

For free response,

$$u(t) = 0$$

so the governing equation becomes

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = 0$$

Assume a solution of the form

$$x(t) = Ce^{st}$$

Then the characteristic equation is

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

Natural Frequency and Damping Ratio

Characteristic roots:

$$s = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

Interpretation:

- $\boldsymbol{\omega_n}$: determines the natural time scale of oscillation
- $\boldsymbol{\zeta}$: determines how strongly the motion is damped

Together, ω_n and ζ determine:

- whether the system oscillates
- how fast the oscillation decays
- how quickly the response settles

Three Damping Cases

Depending on the damping ratio ζ:

Underdamped: $\zeta < 1$

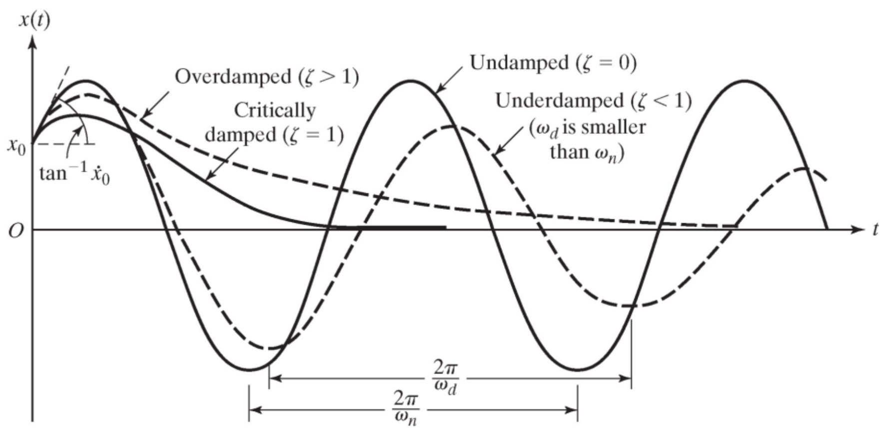
- oscillatory response
- decaying amplitude

Critically damped: $\zeta = 1$

- fastest non-oscillatory response

Overdamped: $\zeta > 1$

- non-oscillatory response
- slower than critical damping



Underdamped Free Response

For $\zeta < 1$, the roots are complex:

$$s_{1,2} = -\zeta\omega_n \pm j\omega_d$$

where

$$\omega_d = \omega_n\sqrt{1 - \zeta^2}$$

Hence, the free response takes the form

$$x(t) = e^{-\zeta\omega_n t}(C_1 \cos \omega_d\, t + C_2 \sin \omega_d\, t)$$

or equivalently

$$x(t) = Ae^{-\zeta\omega_n t}\sin(\omega_d t + \phi)$$

Damped Frequency and Exponential Envelope

Underdamped response:

$$x(t) = Ae^{-\zeta\omega_n t}\sin(\omega_d t + \phi)$$

Key interpretations:

- $\omega_d = \omega_n\sqrt{1-\zeta^2}$: **damped oscillation frequency**
- $e^{-\zeta\omega_n t}$: **exponential decay envelope**

Observations:

- larger $\zeta \rightarrow$ faster decay
- $\omega_d < \omega_n$

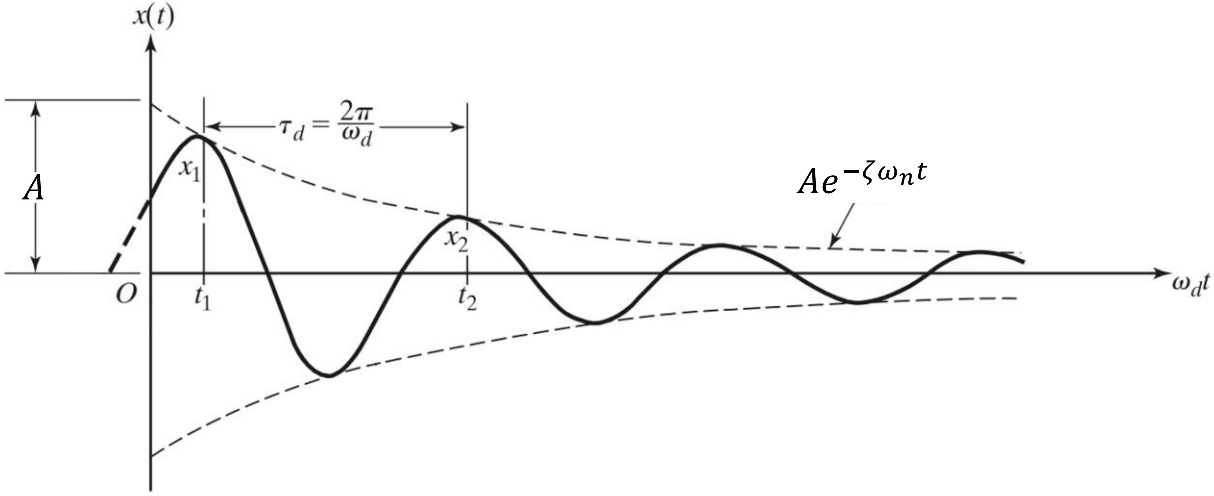
Free Response of an Underdamped System

Underdamped response:

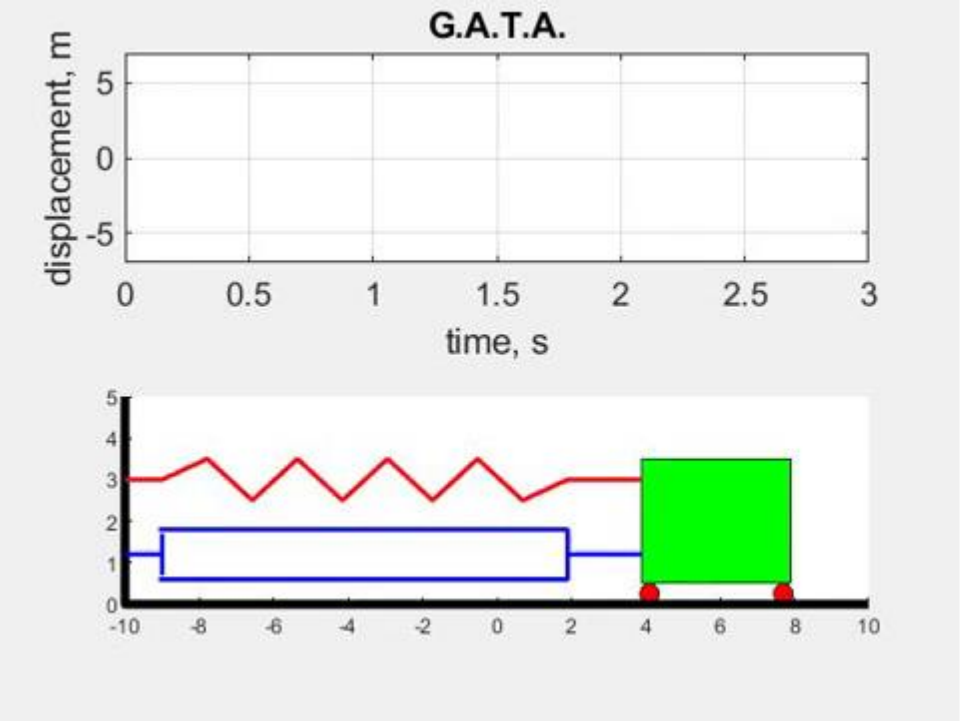
$$x(t) = Ae^{-\zeta\omega_n t}\sin(\omega_d t + \phi)$$

Envelope curves:

$$x(t) = \pm Ae^{-\zeta\omega_n t}$$



Free Response of an Underdamped System



https://www.youtube.com/watch?v=RM6DZD_-sO0

Critically Damped Free Response

For $\zeta = 1$, the characteristic equation becomes

$$s^2 + 2\omega_n s + \omega_n^2 = 0$$

The characteristic roots are repeated:

$$s_1 = s_2 = -\omega_n$$

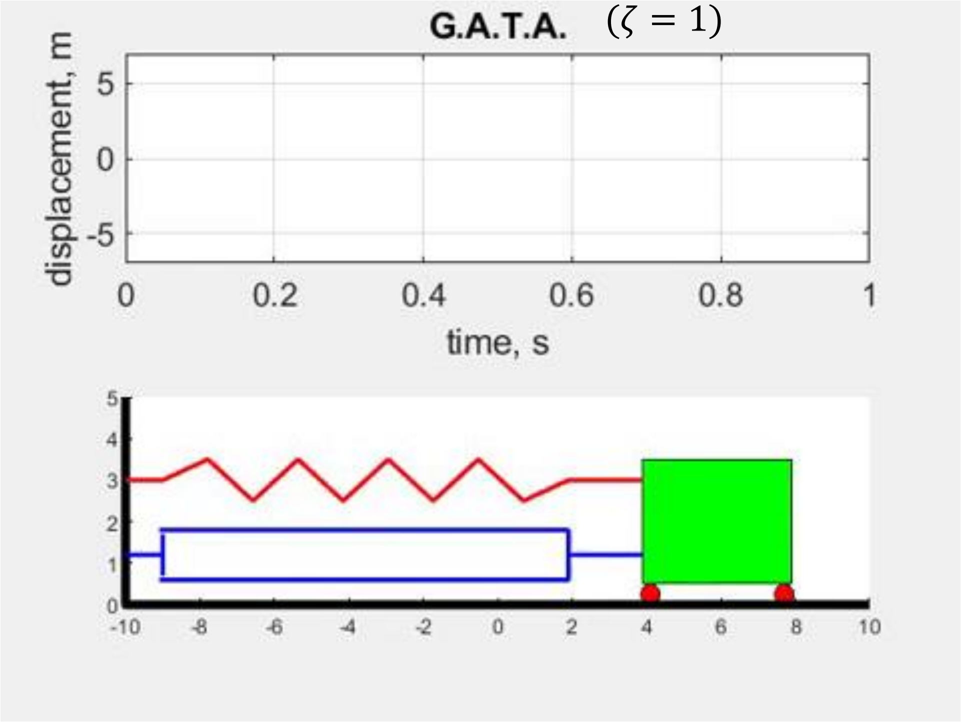
Hence, the general solution is

$$x(t) = (C_1 + C_2 t)e^{-\omega_n t}$$

Response characteristics:

- non-oscillatory
- returns to equilibrium without overshoot
- fastest free response without oscillation

Critically Damped Free Response



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Overdamped Free Response

For $\zeta > 1$, the characteristic roots are real and distinct:

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

Hence, the general solution is

$$x(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t}$$

where s_1 and s_2 are both real and negative.

Response characteristics:

- non-oscillatory
- no overshoot
- slower return to equilibrium than critical damping

Free Response of a Second Order System

Example 1

Consider the free response of the system

$$\ddot{x} + 3\dot{x} + 36x = 0$$

with initial conditions

$$x(0) = 1, \dot{x}(0) = 0$$

Determine:

1. the natural frequency ω_n
2. the damping ratio ζ
3. the type of damping
4. the response of the system

Example 1

Example 1

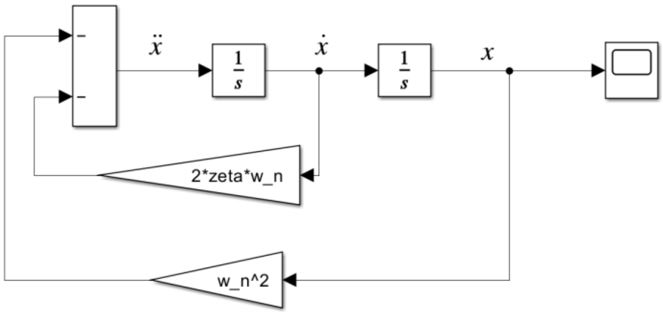
Simulation Check: Free Response

Analytical model:

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = 0$$

Example parameters:

- $\omega_n = 5\ rad/s$
- $\zeta = 0.15, 1.0, 1.5$
- $x(0) = 1$
- $\dot{x}(0) = 0$



Simulink model of a second-order system.

Questions:

- Does the simulation show oscillation?
- Does the amplitude decay exponentially?
- Is the oscillation frequency lower than ω_n?

From Free Response to Step Response

So far:

- response due to **initial conditions only**
- no external input

Next:

- response due to a **step input**
- total response = **transient part + steady-state part**

Questions:

- Will the system still oscillate?
- Will overshoot occur?
- How does damping affect the response?

Step Response of a Second-Order System

Consider the normalized second-order system:

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = b\,u(t)$$

For a step input,

$$u(t) = U_0$$

The response now contains:

- a **transient part**
- a **steady-state part**

Key question:
How does the system move from its initial state to its final value?

Underdamped Step Response

For $\zeta < 1$, the step response is oscillatory and may overshoot.

A standard form is:

$$x(t) = x(\infty)\left[1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin(\omega_d t + \phi)\right]$$

where

$$\omega_d = \omega_n\sqrt{1-\zeta^2}$$

Characteristics:

- oscillatory transient
- overshoot may occur
- amplitude gradually decays
- final value is reached as $t \to \infty$

Critically Damped Step Response

For $\zeta = 1$, the step response is non-oscillatory and takes the form

$$x(t) = x(\infty)[1 - (1 + \omega_n t)e^{-\omega_n t}]$$

Response characteristics:

- no oscillation
- no overshoot
- fastest approach to the final value without oscillation

Overdamped Step Response

For $\zeta > 1$, the step response is non-oscillatory and can be written as

$$x(t) = x(\infty) + C_1 e^{s_1 t} + C_2 e^{s_2 t}$$

where

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

and both roots are real and negative.

Response characteristics:

- no oscillation
- no overshoot
- slower approach to the final value than critical damping

Effect of Damping on Step Response

As damping ratio changes:

Small ζ

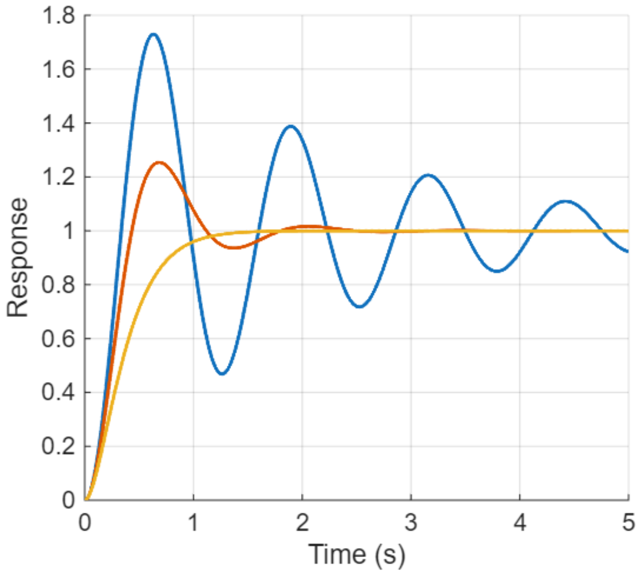
- more oscillation
- larger overshoot
- longer time before settling

Moderate ζ

- less oscillation
- reduced overshoot

Large ζ

- little or no overshoot
- slower but smoother approach to final value



Step Response of a Second-Order System for Different Damping Ratios.

Observation:

Damping strongly shapes the transient response.

Example 2

Consider the system

$$\ddot{x} + 12\dot{x} + 25x = 25u(t)$$

For a unit-step input,

$$u(t) = 1$$

Determine:

1. the natural frequency ω_n
2. the damping ratio ζ
3. the type of damping
4. whether overshoot is expected
5. the steady-state response

Example 2

Step Response of a Second Order System

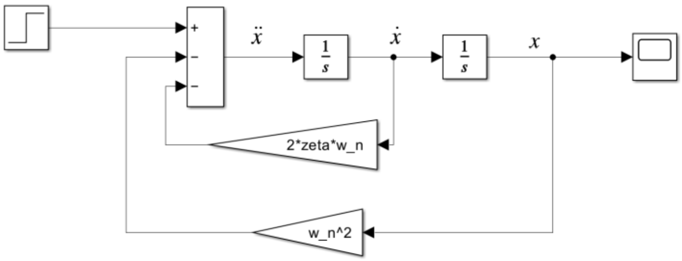
Example 2

Simulation Check: Step Response

Example parameters:

- $\omega_n = 5$ rad/s
- compare $\zeta = 0.1,\ 0.4,\ 1.0$

Apply a unit step input.



Simulink model of a second-order system.

Observe:

- overshoot
- oscillation
- settling trend
- final steady-state value

Question:

How does increasing ζ change the response shape?

Logarithmic Decrement

For an underdamped response:

- oscillation peaks decrease over time
- the decay contains information about damping

In practical work, damping ratio can be estimated from successive peaks using **logarithmic decrement**.

This connects:

- analytical response
- simulated response
- measured experimental response

Decay of Oscillation in an Underdamped System

Underdamped free response:

$$x(t) = Ae^{-\zeta\omega_n t}\sin(\omega_d t + \phi)$$

- The system oscillates after release
- The amplitude decays exponentially with time

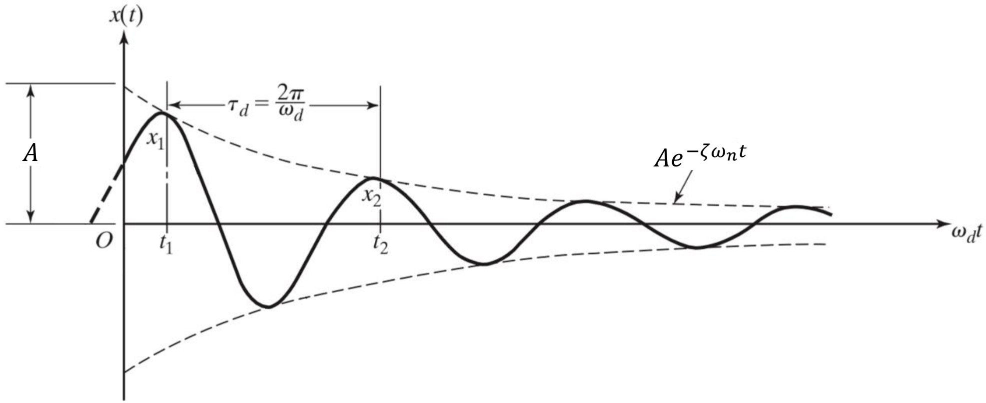
Key observation:
The decay of peak amplitudes can be used to estimate system damping.

Logarithmic Decrement

Logarithmic decrement:

$$\delta = \ln\left(\frac{x_1}{x_2}\right)$$

- δ measures how quickly the oscillation amplitude decays
- Larger δ means heavier damping
- Smaller δ means lighter damping



Successive peaks from the same decay envelope.

Logarithmic Decrement Using Several Cycles

If two peaks are separated by k complete cycles:

$$\delta = \frac{1}{k}\ln\left(\frac{x_1}{x_{k+1}}\right)$$

- More robust for lightly damped systems
- Reduces sensitivity to measurement noise
- Useful when adjacent peaks are very similar

Estimating the Damping Ratio

Relation between logarithmic decrement and damping ratio:

$$\delta = \frac{2\pi\zeta}{\sqrt{1-\zeta^2}}$$

Hence,

$$\zeta = \frac{\delta}{\sqrt{4\pi^2+\delta^2}}$$

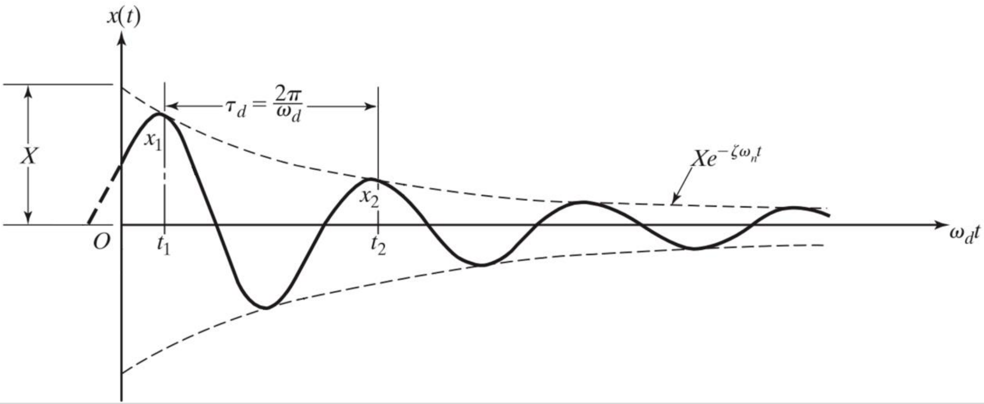
For lightly damped systems ($\zeta \ll 1$):

$$\zeta \approx \frac{\delta}{2\pi}$$

Worked Example: Estimating the Damping Ratio

Example

Suppose the amplitude drops to 25% after 2 cycles:



Review

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4. Link to damping-ratio estimation